

HE2001 Microeconomics II

Midterm Practice 2 (Solutions)

Academic Year 2025/2026, Semester 2

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Overview

This practice paper covers core tools from consumer theory and asset pricing:

- revealed preference (R , P , R^*), WARP and GARP;
- labour–leisure choice and Slutsky decomposition under a wage change;
- present value, NPV, perpetuities, and arbitrage logic;
- FOSD/SSD for discrete contingent consumption bundles;
- state prices, no-arbitrage pricing, and replication;
- Marshallian demand and Slutsky decomposition for Cobb–Douglas-type preferences;
- intertemporal choice with borrowing/saving;
- optimal state-contingent consumption under expected utility.

Instructions

- This exam consists of 5 short questions (8 marks each) and 3 long questions (20 marks each).
- Total marks: 100.
- Show all working clearly. Answers without justification may receive zero marks.
- The notation (x_1, x_2) denotes a bundle of goods 1 and 2.
- The notation (p_1, p_2) denotes prices of goods 1 and 2.

Part I

Short Questions

Question 1 (8 marks)

Alice maximizes a strictly increasing utility function. You observe the following choices:

- At prices $p^1 = (2, 1)$, she buys $x^1 = (4, 2)$
- At prices $p^2 = (1, 2)$, she buys $x^2 = (2, 4)$
- At prices $p^3 = (1, 1)$, she buys $x^3 = (3, 3)$

- (2 marks) Compute expenditure in each observation.
- (3 marks) Determine all nontrivial direct revealed preference relations and strict direct revealed preference relations among x^1, x^2, x^3 .
- (3 marks) Does the data satisfy WARP? Does it satisfy GARP? Clearly justify using R , P , and (if needed) R^* .

Solution

- Let $m^t = p^t \cdot x^t$.

$$m^1 = (2, 1) \cdot (4, 2) = 10,$$

$$m^2 = (1, 2) \cdot (2, 4) = 10,$$

$$m^3 = (1, 1) \cdot (3, 3) = 6.$$

Hence

$$m^1 = 10, \quad m^2 = 10, \quad m^3 = 6.$$

- Recall

$$x^t R x^s \iff p^t \cdot x^s \leq p^t \cdot x^t, \quad x^t P x^s \iff p^t \cdot x^s < p^t \cdot x^t.$$

Compute cross-costs:

	x^1	x^2	x^3
$p^1 \cdot (\cdot)$	10	8	9
$p^2 \cdot (\cdot)$	8	10	9
$p^3 \cdot (\cdot)$	6	6	6

Therefore the nontrivial direct revealed preference relations are

$$x^1 R x^2, x^1 R x^3, x^2 R x^1, x^2 R x^3, x^3 R x^1, x^3 R x^2.$$

The strict direct revealed preference relations are those with strict inequality:

$$x^1 P x^2, x^1 P x^3, x^2 P x^1, x^2 P x^3.$$

At observation 3, x^1 and x^2 are only weakly affordable (equal cost), so $x^3 R x^1$ and $x^3 R x^2$ but not strict.

- (c) **WARP:** WARP is violated if there exist $t \neq s$ such that $x^t R x^s$ and $x^s P x^t$. Here,

$$x^1 R x^2 \quad \text{and} \quad x^2 P x^1,$$

so WARP fails. (Equivalently, $x^2 R x^1$ and $x^1 P x^2$ also violates WARP.) Thus

WARP is violated.

GARP: Since WARP fails, GARP also fails (a WARP violation is a special case of a GARP violation). Explicitly, because $x^1 R x^2$ (hence $x^1 R^* x^2$) and

$$p^2 \cdot x^1 = 8 < 10 = p^2 \cdot x^2,$$

GARP is violated. Hence

GARP is violated.

Question 2 (8 marks)

A worker has utility

$$U(c, \ell) = c^{1/2} \ell^{1/2},$$

where c is consumption and ℓ is leisure (hours). The worker has 16 hours per day, wage $w > 0$, and non-labour income $y = 40$.

- (a) (3 marks) Write the budget constraint and derive the optimal labour supply function $L^*(w)$ (hours worked), assuming an interior solution.
- (b) (2 marks) Compute optimal labour supply and consumption when $w = 10$.
- (c) (3 marks) The wage rises to $w' = 20$. Using the Slutsky method (compensating non-labour income so the original bundle remains affordable), compute the substitution effect and income effect on labour supply.

Solution

Let total time be $T = 16$, labour hours be L , and leisure be $\ell = T - L = 16 - L$. Consumption is

$$c = wL + y = w(16 - \ell) + 40.$$

Equivalently, the budget in (c, ℓ) -space is

$$c + w\ell = 16w + 40.$$

- (a) Since $U(c, \ell) = c^{1/2}\ell^{1/2}$ is Cobb–Douglas, an interior optimum spends half of “full income” on each good in value terms:

$$c^* = \frac{16w + 40}{2}, \quad w\ell^* = \frac{16w + 40}{2}.$$

Hence

$$\ell^* = \frac{16w + 40}{2w} = 8 + \frac{20}{w}.$$

Therefore labour supply is

$$L^*(w) = 16 - \ell^* = 16 - \left(8 + \frac{20}{w}\right) = 8 - \frac{20}{w}.$$

So (for an interior solution)

$$L^*(w) = 8 - \frac{20}{w}.$$

(Interior requires $0 < L^*(w) < 16$, which holds in particular for $w = 10, 20$.)

- (b) At $w = 10$,

$$L^*(10) = 8 - \frac{20}{10} = 6.$$

Then leisure is $\ell^* = 10$, and consumption is

$$c^* = 10 \cdot 6 + 40 = 100.$$

Hence

$$L^* = 6, \quad c^* = 100.$$

- (c) Original optimum at $w = 10$ is $(L^0, c^0) = (6, 100)$, equivalently $\ell^0 = 10$. When wage rises to $w' = 20$, choose compensated non-labour income y^s so that the *original bundle* remains affordable:

$$c^0 = w'L^0 + y^s.$$

Thus

$$100 = 20 \cdot 6 + y^s \Rightarrow y^s = -20.$$

So the compensated labour supply at $w' = 20$ is

$$L^s = 8 - \frac{y^s}{2w'} = 8 - \frac{-20}{40} = 8.5,$$

using the general formula $L^*(w, y) = 8 - \frac{y}{2w}$. (Equivalently, from $L^*(w) = 8 - 20/w$ after replacing 40 by y .)

The actual post-change labour supply (with actual non-labour income $y = 40$) is

$$L^1 = 8 - \frac{20}{20} = 7.$$

Therefore:

$$\Delta L^s = L^s - L^0 = 8.5 - 6 = 2.5, \quad \Delta L^n = L^1 - L^s = 7 - 8.5 = -1.5.$$

Hence

$$\boxed{\Delta L^s = +2.5, \quad \Delta L^n = -1.5.}$$

The substitution effect increases labour supply; the income effect reduces labour supply.

Question 3 (8 marks)

Consider a three-period economy ($t = 0, 1, 2$) with interest rate $r = 0.25$.

- (a) (2 marks) Find the present value at $t = 0$ of the cashflow stream $(0, 125, 156.25)$.
- (b) (3 marks) A project costs 210 today and pays $(0, 50, 250)$ in periods $(0, 1, 2)$ respectively (so the $t = 0$ component is the cost). Compute its NPV and state whether it should be accepted.
- (c) (3 marks) A perpetuity pays 30 every period starting from $t = 1$. If its market price is 130, determine whether an arbitrage opportunity exists (assume short selling is allowed) and describe a valid arbitrage strategy.

Solution

- (a) Present value:

$$PV = \frac{125}{1.25} + \frac{156.25}{(1.25)^2} = 100 + 100 = 200.$$

So

$$\boxed{PV = 200.}$$

- (b) The project has cashflow $(-210, 50, 250)$. Its present value of payoffs (excluding the initial cost) is

$$\frac{50}{1.25} + \frac{250}{(1.25)^2} = 40 + 160 = 200.$$

Thus

$$NPV = -210 + 200 = -10.$$

Hence

$$\boxed{NPV = -10, \text{ so the project should be rejected.}}$$

- (c) The no-arbitrage price of a perpetuity paying 30 from $t = 1$ onward is

$$P^* = \frac{30}{r} = \frac{30}{0.25} = 120.$$

The market price is $130 > 120$, so the perpetuity is overpriced.

An arbitrage strategy is:

- short 1 unit of the perpetuity (receive 130 today);
- use 120 to buy a self-financing risk-free fund that generates 30 each period forever (equivalently, invest 120 at 25% and pay interest only);
- pocket the remaining 10 today.

The future liability from the short perpetuity is exactly covered by the 30 cashflow each period from the fund. Thus

Yes, arbitrage exists: short the perpetuity and lock in 10 today.

Question 4 (8 marks)

There are four equally likely states. Consider two contingent consumption bundles (written in ascending order of states):

$$x = (2, 4, 8, 10), \quad y = (1, 5, 7, 11).$$

- (2 marks) Compute $\mathbb{E}[x]$ and $\mathbb{E}[y]$.
- (3 marks) Does x first-order stochastically dominate y ? Does y first-order stochastically dominate x ? Justify using CDF comparisons at the relevant support points.
- (3 marks) Determine whether x second-order stochastically dominates y (or vice versa). Use an appropriate discrete SSD test.

Solution

- Since all states are equally likely,

$$\mathbb{E}[x] = \frac{2 + 4 + 8 + 10}{4} = \frac{24}{4} = 6, \quad \mathbb{E}[y] = \frac{1 + 5 + 7 + 11}{4} = \frac{24}{4} = 6.$$

Hence

$$\mathbb{E}[x] = \mathbb{E}[y] = 6.$$

- FOSD test via CDFs.** Let F_x and F_y be the CDFs of the equally weighted lotteries induced by x and y . For FOSD, we need one CDF to be everywhere weakly below the other.

Check a few support points where ordering can change:

- At $c = 4.5$,

$$F_x(4.5) = \Pr(x \leq 4.5) = \frac{2}{4} = \frac{1}{2}, \quad F_y(4.5) = \Pr(y \leq 4.5) = \frac{1}{4}.$$

So $F_x(4.5) > F_y(4.5)$.

- At $c = 7.5$,

$$F_x(7.5) = \Pr(x \leq 7.5) = \frac{2}{4} = \frac{1}{2}, \quad F_y(7.5) = \Pr(y \leq 7.5) = \frac{3}{4}.$$

So $F_x(7.5) < F_y(7.5)$.

The CDFs cross, so neither lottery FOSD-dominates the other. Thus

Neither x FOSD y nor y FOSD x .

- (c) **SSD test (equal probabilities, sorted outcomes).** For equally likely outcomes sorted in ascending order, x SSD-dominates y iff the cumulative sums of the ordered outcomes satisfy

$$\sum_{i=1}^k x_{(i)} \geq \sum_{i=1}^k y_{(i)} \quad (k = 1, 2, 3),$$

and total sums are equal. Here the outcomes are already sorted:

For x : $2, 2 + 4 = 6, 2 + 4 + 8 = 14, 24$,

For y : $1, 1 + 5 = 6, 1 + 5 + 7 = 13, 24$.

Hence

$$2 \geq 1, \quad 6 = 6, \quad 14 \geq 13,$$

and the totals are equal ($24 = 24$). Therefore

x second-order stochastically dominates y .

(Equivalently, all risk-averse expected-utility consumers weakly prefer x to y .)

Question 5 (8 marks)

There are two states of nature, state 1 and state 2. Two assets are traded:

- Asset A pays $(4, 0)$ and costs 1.00
 - Asset B pays $(1, 2)$ and costs 1.25
- (a) (3 marks) Find the state price vector (q_1, q_2) . Also find the price and gross return of a risk-free asset paying $(1, 1)$.
- (b) (2 marks) What is the no-arbitrage price of the contingent claim paying $(3, 5)$?
- (c) (3 marks) A new asset pays $(2, 1)$. What is its no-arbitrage price? If its market price is 0.90, describe an arbitrage strategy.

Solution

Let $q = (q_1, q_2)$ denote the state price vector. No-arbitrage implies each asset price equals q dot payoff.

(a) From Asset A:

$$4q_1 + 0q_2 = 1.00 \Rightarrow q_1 = 0.25.$$

From Asset B:

$$q_1 + 2q_2 = 1.25 \Rightarrow 0.25 + 2q_2 = 1.25 \Rightarrow q_2 = 0.50.$$

Therefore

$$(q_1, q_2) = (0.25, 0.50).$$

A risk-free payoff $(1, 1)$ has price

$$p_f = q_1 + q_2 = 0.75.$$

Hence the gross risk-free return is

$$R_f = \frac{1}{p_f} = \frac{1}{0.75} = \frac{4}{3}.$$

So

$$p_f = 0.75, \quad R_f = \frac{4}{3}.$$

(b) Price of claim $(3, 5)$:

$$p = 3q_1 + 5q_2 = 3(0.25) + 5(0.50) = 0.75 + 2.5 = 3.25.$$

Hence

$$p = 3.25.$$

(c) No-arbitrage price of payoff $(2, 1)$ is

$$p^* = 2q_1 + q_2 = 2(0.25) + 0.50 = 1.00.$$

So

$$p^* = 1.00.$$

If the market price is 0.90, the asset is underpriced. A valid arbitrage strategy is:

- buy 1 unit of the new asset for 0.90;
- short the replicating portfolio for payoff $(2, 1)$ (which costs 1.00).

One replication of $(2, 1)$ using A and B is obtained by solving

$$\theta_A(4, 0) + \theta_B(1, 2) = (2, 1).$$

From state 2: $2\theta_B = 1 \Rightarrow \theta_B = \frac{1}{2}$. From state 1: $4\theta_A + \frac{1}{2} = 2 \Rightarrow \theta_A = \frac{3}{8}$. Thus the replica costs

$$\frac{3}{8}(1.00) + \frac{1}{2}(1.25) = 0.375 + 0.625 = 1.00.$$

Shorting this replica generates 1.00 today; buying the underpriced new asset costs 0.90; future payoffs cancel exactly. Initial arbitrage profit:

$$1.00 - 0.90 = \boxed{0.10}.$$

Part II

Long Questions

Question 6 (20 marks)

Consider a consumer with utility function

$$U(x_1, x_2) = \ln x_1 + 2 \ln x_2,$$

with $x_1 > 0$ and $x_2 > 0$.

- (4 marks) Compute MU_1 , MU_2 , and the marginal rate of substitution $MRS_{1,2}$. State the interior first-order condition for utility maximization at prices (p_1, p_2) .
- (5 marks) Derive the Marshallian demand functions $x_1(p_1, p_2, m)$ and $x_2(p_1, p_2, m)$.
- (4 marks) The consumer has endowment $\omega = (9, 6)$ and faces prices $(p_1, p_2) = (2, 3)$. Compute the value of the endowment, the optimal consumption bundle, and whether the consumer is a net buyer or seller of each good.
- (3 marks) Verify Walras' Law at the bundle obtained in part (c).
- (4 marks) Holding nominal income fixed at the value from part (c), suppose the price of good 1 rises from $p_1 = 2$ to $p'_1 = 3$ (with $p_2 = 3$ unchanged). Using the Slutsky decomposition for good 1, compute the substitution effect and income effect, and state whether good 1 is normal.

Solution

- Marginal utilities:

$$MU_1 = \frac{\partial U}{\partial x_1} = \frac{1}{x_1}, \quad MU_2 = \frac{\partial U}{\partial x_2} = \frac{2}{x_2}.$$

Hence the marginal rate of substitution of good 1 for good 2 is

$$MRS_{1,2} = \frac{MU_1}{MU_2} = \frac{1/x_1}{2/x_2} = \frac{x_2}{2x_1}.$$

At an interior optimum,

$$MRS_{1,2} = \frac{p_1}{p_2} \iff \frac{x_2}{2x_1} = \frac{p_1}{p_2}.$$

So

$$\boxed{MU_1 = \frac{1}{x_1}, \quad MU_2 = \frac{2}{x_2}, \quad MRS_{1,2} = \frac{x_2}{2x_1}, \text{ and } \frac{x_2}{2x_1} = \frac{p_1}{p_2}.}$$

(b) Solve

$$\max_{x_1, x_2 > 0} \ln x_1 + 2 \ln x_2 \quad \text{s.t.} \quad p_1 x_1 + p_2 x_2 = m.$$

Using a Lagrangian,

$$\mathcal{L} = \ln x_1 + 2 \ln x_2 + \lambda(m - p_1 x_1 - p_2 x_2).$$

FOCs:

$$\frac{1}{x_1} = \lambda p_1, \quad \frac{2}{x_2} = \lambda p_2.$$

Thus

$$x_1 = \frac{1}{\lambda p_1}, \quad x_2 = \frac{2}{\lambda p_2}.$$

Substitute into the budget:

$$p_1 \left(\frac{1}{\lambda p_1} \right) + p_2 \left(\frac{2}{\lambda p_2} \right) = m \Rightarrow \frac{3}{\lambda} = m \Rightarrow \lambda = \frac{3}{m}.$$

Therefore

$$x_1(p_1, p_2, m) = \frac{m}{3p_1}, \quad x_2(p_1, p_2, m) = \frac{2m}{3p_2}.$$

(c) Endowment value at prices (2, 3):

$$m = p \cdot \omega = 2 \cdot 9 + 3 \cdot 6 = 18 + 18 = 36.$$

Using the demands,

$$x_1^* = \frac{36}{3 \cdot 2} = 6, \quad x_2^* = \frac{2 \cdot 36}{3 \cdot 3} = 8.$$

Hence

$$m = 36, \quad x^* = (6, 8).$$

Net trades relative to endowment $\omega = (9, 6)$:

$$x^* - \omega = (6 - 9, 8 - 6) = (-3, 2).$$

So the consumer is a net **seller** of good 1 (sells 3 units) and a net **buyer** of good 2 (buys 2 units).

(d) Walras' Law requires

$$p \cdot x(p, m) = m.$$

At the bundle from part (c),

$$p \cdot x^* = 2 \cdot 6 + 3 \cdot 8 = 12 + 24 = 36 = m.$$

Thus

$$p \cdot x^* = p \cdot \omega = 36.$$

Walras' Law holds.

(e) Original prices: $p = (2, 3)$, income $m = 36$. Old Marshallian demand for good 1:

$$x_1^0 = \frac{36}{3 \cdot 2} = 6.$$

New prices: $p' = (3, 3)$ with income still fixed at $m = 36$. New Marshallian demand:

$$x_1^1 = \frac{36}{3 \cdot 3} = 4.$$

Total change:

$$\Delta x_1 = x_1^1 - x_1^0 = 4 - 6 = -2.$$

For the Slutsky decomposition, compensate income so that the original bundle $x^0 = (6, 8)$ is affordable at new prices:

$$m^s = p' \cdot x^0 = 3 \cdot 6 + 3 \cdot 8 = 42.$$

Compensated (Slutsky) demand for good 1 at (p', m^s) is

$$x_1^s = \frac{42}{3 \cdot 3} = \frac{42}{9} = \frac{14}{3}.$$

Hence

$$\Delta x_1^s = x_1^s - x_1^0 = \frac{14}{3} - 6 = -\frac{4}{3},$$

and

$$\Delta x_1^n = x_1^1 - x_1^s = 4 - \frac{14}{3} = -\frac{2}{3}.$$

So

$$\boxed{\Delta x_1^s = -\frac{4}{3}, \quad \Delta x_1^n = -\frac{2}{3}.$$

Because $x_1 = \frac{m}{3p_1}$ increases with income m , good 1 is normal. Equivalently, the income effect above has the normal-good sign (a fall in real income reduces demand for good 1).

Question 7 (20 marks)

Consider a two-period intertemporal consumption problem with utility

$$U(x_0, x_1) = \ln x_0 + \ln x_1,$$

where x_0 is period-0 consumption and x_1 is period-1 consumption. The consumer can borrow or save at interest rate r .

- (3 marks) Write the intertemporal budget constraint in present-value form and in future-value form.
- (5 marks) Let m denote the present value of lifetime income. Derive the optimal demands $x_0^*(r, m)$ and $x_1^*(r, m)$.
- (4 marks) Suppose the income stream is $(y_0, y_1) = (18, 30)$ and $r = 0.25$. Compute the present value of income, the optimal consumption bundle, and whether the consumer is a net borrower or net saver in period 0 (and by how much).
- (4 marks) Verify that the optimal bundle from part (c) satisfies the budget constraint exactly.
- (4 marks) Now suppose the interest rate rises to $r' = 0.50$. Using Slutsky decomposition for period-0 consumption x_0 , compute the Slutsky compensated present value m^s , the substitution effect, and the income effect. Is period-0 consumption normal?

Solution

- Let lifetime income be (y_0, y_1) . The present-value budget constraint is

$$x_0 + \frac{x_1}{1+r} = y_0 + \frac{y_1}{1+r}.$$

If we define present value income

$$m := y_0 + \frac{y_1}{1+r},$$

then the PV budget is simply

$$\boxed{x_0 + \frac{x_1}{1+r} = m.}$$

Multiplying by $(1+r)$ gives the future-value form:

$$\boxed{(1+r)x_0 + x_1 = (1+r)y_0 + y_1.}$$

(b) Solve

$$\max_{x_0, x_1 > 0} \ln x_0 + \ln x_1 \quad \text{s.t.} \quad x_0 + \frac{x_1}{1+r} = m.$$

Lagrangian:

$$\mathcal{L} = \ln x_0 + \ln x_1 + \lambda \left(m - x_0 - \frac{x_1}{1+r} \right).$$

FOCs:

$$\frac{1}{x_0} = \lambda, \quad \frac{1}{x_1} = \frac{\lambda}{1+r}.$$

Hence

$$x_1 = (1+r)x_0.$$

Substitute into the budget:

$$x_0 + \frac{(1+r)x_0}{1+r} = 2x_0 = m \Rightarrow x_0^* = \frac{m}{2}.$$

Then

$$x_1^* = (1+r)x_0^* = \frac{(1+r)m}{2}.$$

Therefore

$$x_0^*(r, m) = \frac{m}{2}, \quad x_1^*(r, m) = \frac{(1+r)m}{2}.$$

(c) With $(y_0, y_1) = (18, 30)$ and $r = 0.25$,

$$m = 18 + \frac{30}{1.25} = 18 + 24 = 42.$$

Thus optimal consumption is

$$x_0^* = \frac{42}{2} = 21, \quad x_1^* = \frac{1.25 \cdot 42}{2} = 26.25 = \frac{105}{4}.$$

Hence

$$m = 42, \quad (x_0^*, x_1^*) = (21, 26.25).$$

Since $x_0^* = 21 > y_0 = 18$, the consumer consumes more than current income in period 0 and therefore **borrow**s:

$$\text{Borrowing in period 0} = x_0^* - y_0 = 21 - 18 = 3.$$

(d) Verify the PV budget:

$$x_0^* + \frac{x_1^*}{1+r} = 21 + \frac{26.25}{1.25} = 21 + 21 = 42 = m.$$

So the constraint holds exactly. Equivalently, in FV form:

$$1.25(21) + 26.25 = 26.25 + 26.25 = 52.5,$$

while

$$1.25(18) + 30 = 22.5 + 30 = 52.5.$$

Hence

$$\boxed{\text{Budget constraint satisfied exactly.}}$$

(e) Old optimum (from part (c)) at $r = 0.25$ is

$$(x_0^0, x_1^0) = \left(21, \frac{105}{4}\right).$$

When the interest rate rises to $r' = 0.50$, the *actual* present value of income becomes

$$m' = 18 + \frac{30}{1.5} = 18 + 20 = 38.$$

Thus actual new period-0 demand is

$$x_0^1 = \frac{m'}{2} = 19.$$

For the Slutsky decomposition, choose compensated present value m^s so that the old bundle remains affordable at the new interest rate:

$$m^s = x_0^0 + \frac{x_1^0}{1 + r'} = 21 + \frac{105/4}{1.5} = 21 + \frac{105}{6} = 21 + 17.5 = 38.5 = \frac{77}{2}.$$

So the compensated demand is

$$x_0^s = \frac{m^s}{2} = \frac{77}{4} = 19.25.$$

Therefore

$$\Delta x_0^s = x_0^s - x_0^0 = \frac{77}{4} - 21 = -\frac{7}{4} = -1.75,$$

and

$$\Delta x_0^n = x_0^1 - x_0^s = 19 - 19.25 = -\frac{1}{4} = -0.25.$$

Hence

$$\boxed{m^s = 38.5, \quad \Delta x_0^s = -1.75, \quad \Delta x_0^n = -0.25.}$$

Since $x_0^*(r, m) = m/2$ is increasing in m , period-0 consumption is normal.

Question 8 (20 marks)

There are three states of nature $s = 1, 2, 3$. Three traded assets have the following state payoffs and prices:

Asset	$s = 1$	$s = 2$	$s = 3$	Price
1	2	0	0	0.40
2	1	1	0	0.50
3	1	1	1	0.90

A consumer has expected utility

$$U(x_1, x_2, x_3) = \frac{1}{2} \ln x_1 + \frac{1}{3} \ln x_2 + \frac{1}{6} \ln x_3,$$

and endowment $\omega = (4, 2, 1)$.

- (5 marks) Find the state price vector (q_1, q_2, q_3) . Then compute the price and gross return of the risk-free payoff $(1, 1, 1)$.
- (3 marks) Compute the value of the endowment and the no-arbitrage price of the contingent claim $d = (3, 1, 2)$.
- (6 marks) Using the state-price budget constraint, solve for the consumer's optimal contingent consumption bundle (x_1^*, x_2^*, x_3^*) .
- (3 marks) Find a portfolio $(\theta_1, \theta_2, \theta_3)$ of the traded assets that replicates the optimal bundle from part (c), and verify that its cost equals the state-price cost of the bundle.
- (3 marks) Suppose the claim $d = (3, 1, 2)$ from part (b) is instead traded at price 1.80. Is there an arbitrage opportunity? If yes, describe one and compute the initial arbitrage profit.

Solution

Let $q = (q_1, q_2, q_3)$ be the state price vector.

- Use no-arbitrage pricing on the three traded assets:

$$2q_1 = 0.40, \quad q_1 + q_2 = 0.50, \quad q_1 + q_2 + q_3 = 0.90.$$

Solve sequentially:

$$q_1 = 0.20, \quad q_2 = 0.50 - 0.20 = 0.30, \quad q_3 = 0.90 - 0.50 = 0.40.$$

Hence

$$(q_1, q_2, q_3) = (0.20, 0.30, 0.40).$$

The risk-free payoff $(1, 1, 1)$ has price

$$p_f = q_1 + q_2 + q_3 = 0.90.$$

Therefore its gross return is

$$R_f = \frac{1}{p_f} = \frac{1}{0.90} = \frac{10}{9}.$$

So

$$p_f = 0.90, \quad R_f = \frac{10}{9}.$$

(b) Endowment value:

$$m = q \cdot \omega = 0.20(4) + 0.30(2) + 0.40(1) = 0.8 + 0.6 + 0.4 = 1.8.$$

So

$$q \cdot \omega = 1.8.$$

The contingent claim $d = (3, 1, 2)$ has no-arbitrage price

$$p_d = q \cdot d = 0.20(3) + 0.30(1) + 0.40(2) = 0.6 + 0.3 + 0.8 = 1.7.$$

Thus

$$p_d^* = 1.70.$$

(c) The state-price budget constraint is

$$0.20x_1 + 0.30x_2 + 0.40x_3 = m = 1.8.$$

Maximize

$$\frac{1}{2} \ln x_1 + \frac{1}{3} \ln x_2 + \frac{1}{6} \ln x_3$$

subject to that budget. Using a Lagrangian,

$$\mathcal{L} = \frac{1}{2} \ln x_1 + \frac{1}{3} \ln x_2 + \frac{1}{6} \ln x_3 + \lambda(1.8 - 0.2x_1 - 0.3x_2 - 0.4x_3).$$

FOCs:

$$\frac{1/2}{x_1} = 0.2\lambda, \quad \frac{1/3}{x_2} = 0.3\lambda, \quad \frac{1/6}{x_3} = 0.4\lambda.$$

Equivalently, state expenditures follow the utility weights (which sum to 1):

$$q_1x_1 = \frac{1}{2}m, \quad q_2x_2 = \frac{1}{3}m, \quad q_3x_3 = \frac{1}{6}m.$$

Since $m = 1.8$,

$$q_1x_1 = 0.9, \quad q_2x_2 = 0.6, \quad q_3x_3 = 0.3.$$

Hence

$$x_1^* = \frac{0.9}{0.2} = 4.5, \quad x_2^* = \frac{0.6}{0.3} = 2, \quad x_3^* = \frac{0.3}{0.4} = 0.75.$$

Therefore

$$(x_1^*, x_2^*, x_3^*) = (4.5, 2, 0.75) = \left(\frac{9}{2}, 2, \frac{3}{4}\right).$$

(d) Let asset holdings be $(\theta_1, \theta_2, \theta_3)$. Their state payoff is

$$\theta_1(2, 0, 0) + \theta_2(1, 1, 0) + \theta_3(1, 1, 1).$$

Set this equal to the target bundle $(\frac{9}{2}, 2, \frac{3}{4})$:

$$\begin{cases} 2\theta_1 + \theta_2 + \theta_3 = \frac{9}{2}, \\ \theta_2 + \theta_3 = 2, \\ \theta_3 = \frac{3}{4}. \end{cases}$$

From the third equation, $\theta_3 = \frac{3}{4}$. Then the second gives

$$\theta_2 = 2 - \frac{3}{4} = \frac{5}{4}.$$

Then the first gives

$$2\theta_1 + \frac{5}{4} + \frac{3}{4} = \frac{9}{2} \Rightarrow 2\theta_1 + 2 = \frac{9}{2} \Rightarrow 2\theta_1 = \frac{5}{2} \Rightarrow \theta_1 = \frac{5}{4}.$$

Thus

$$(\theta_1, \theta_2, \theta_3) = \left(\frac{5}{4}, \frac{5}{4}, \frac{3}{4}\right).$$

Portfolio cost:

$$0.40 \left(\frac{5}{4}\right) + 0.50 \left(\frac{5}{4}\right) + 0.90 \left(\frac{3}{4}\right) = 0.5 + 0.625 + 0.675 = 1.8.$$

State-price cost of the bundle is also

$$q \cdot x^* = 1.8.$$

Hence the costs match:

$$\boxed{\text{portfolio cost} = \text{state-price cost} = 1.8.}$$

(e) From part (b), the no-arbitrage price of $d = (3, 1, 2)$ is 1.70. If it trades at 1.80, it is overpriced, so an arbitrage exists.

One replicating portfolio for d solves

$$\theta_1(2, 0, 0) + \theta_2(1, 1, 0) + \theta_3(1, 1, 1) = (3, 1, 2).$$

Thus

$$\begin{cases} 2\theta_1 + \theta_2 + \theta_3 = 3, \\ \theta_2 + \theta_3 = 1, \\ \theta_3 = 2. \end{cases}$$

So $\theta_3 = 2$, then $\theta_2 = -1$, and then $2\theta_1 - 1 + 2 = 3 \Rightarrow \theta_1 = 1$. Hence the replica is $(\theta_1, \theta_2, \theta_3) = (1, -1, 2)$ and costs

$$1(0.40) + (-1)(0.50) + 2(0.90) = 0.40 - 0.50 + 1.80 = 1.70.$$

Arbitrage strategy:

- short 1 unit of the overpriced claim d (receive 1.80),
- buy its replicating portfolio (pay 1.70).

Initial riskless profit:

$$1.80 - 1.70 = \boxed{0.10}.$$

Future state payoffs cancel exactly.